

Santana et al. 2023 — Reproduction Report

Paper: Santana DJ et al. "A Candida auris-specific adhesin, Scf1, governs surface association, colonization, and virulence." Science 381:1461–1466 (2023). PMC11235122.

Our reproduction: 6 paired-end RNA-Seq libraries (NextSeq 2000, 2×50 bp, reverse-stranded, Ribo-Zero Plus), 3 biological pairs from BioProject PRJNA904261, mapped to C. auris B8441 v3 (GCA_002759435.3) with the IWC `rnaseq-pe` workflow → STAR → featureCounts → DESeq2.

1. SCF1 (B9J08_001458 / v3 B9J08_03708) — central finding

Contrast	Paper claim	Our result	Verdict
AR0382 tnSWI1 vs WT (Fig. 1D)	Most strongly down-regulated gene in the mutant; presented as the headline hit	log2FC = −5.83 , padj = 0.023 — rank 7 / 978 sig. genes by absolute log2FC (top 0.7 %)	Reproduced — direction + magnitude consistent; SCF1 sits alone in the bottom-left of our volcano (<code>figures/ volcano_Fig1D_tnSWI1_vs_WT.pdf</code>)
AR0387 vs AR0382 (Fig. 2D / S5)	~29-fold lower in AR0387 → expected log2FC ≈ −4.86	log2FC = −6.35 , padj = 0.0045 — rank 3 / 1690 sig. genes (top 0.2 %)	Reproduced and slightly stronger than paper's point estimate ; rank-among-DE- genes confirms it is one of the top hits

This is the paper's central transcriptomic claim and it survives independent re-analysis on a different annotation (v3 vs the v2 the authors used).

2. IFF / HYR adhesin family (specificity control)

Paper specifically noted that tnSWI1 disruption does **not** dysregulate other GPI-anchored adhesins, arguing SCF1 is the relevant effector rather than a general adhesin-program collapse.

Gene	tnSWI1 vs WT log2FC (padj)	AR0387 vs AR0382 log2FC (padj)	Paper claim	Verdict
IFF4109 (B9J08_04863)	+0.02 (0.98)	-0.60 (0.0011)	Not dysregulated in tnSWI1; differentially expressed in AR0387	Reproduced both
IFF4892 (B9J08_04852)	-0.06 (0.75)	+0.13 (0.29)	Similar expression to IFF4109 but not required for adhesion	Reproduced (no strong DE in either contrast)
ALS4112 (B9J08_04866)	-0.21 (0.83)	-1.08 (0.022)	Mentioned as DE in AR0387	Reproduced in AR0387; not in tnSWI1 (consistent with paper)

The IFF/HYR/ALS family does not collapse in tnSWI1, exactly as the paper argues. This is the negative-control evidence that SCF1 is the specific lesion.

3. SWI1 itself (B9J08_003460 / v3 B9J08_01319) — transposon-insertion artifact

Paper does not call SWI1 itself differentially expressed in tnSWI1 (the insertion is genomic, not transcriptional). Our result:

Contrast	log2FC	padj	Interpretation
tnSWI1 vs WT	+0.62	2 × 10⁻¹⁸	Reads pile up on the 5' portion upstream of the transposon insertion — classic tn-mutant signature, expected, not a biological effect
AR0387 vs AR0382	-0.40	0.087	Not significant — consistent with AR0387 being WT for SWI1

This is a quality-control signal: the tnSWI1 sample really is tnSWI1.

4. BCY1 (cAMP/PKA regulator)

Paper mentions BCY1 in the context of cAMP/PKA signaling downstream of SCF1. Our DE values are modest:

Contrast	log2FC	padj
tnSWI1 vs WT	-0.20	0.11 (ns)
AR0387 vs AR0382	+0.21	0.034

Weakly DE in AR0387, not in tnSWI1. Paper does not lean heavily on BCY1 transcript abundance — its role is post-translational — so the lack of large DE is **consistent**, not contradictory.

5. Global DE summary

Contrast	Down (lfc<-1, padj<0.05)	Up (lfc>+1, padj<0.05)	Total padj<0.05
AR0382 tnSWI1 vs WT	183	99	978
AR0387 vs AR0382	138	102	1690

The AR0387 contrast has more total DE (clade-level divergence > single-gene insertion), which matches the biological expectation in the paper.

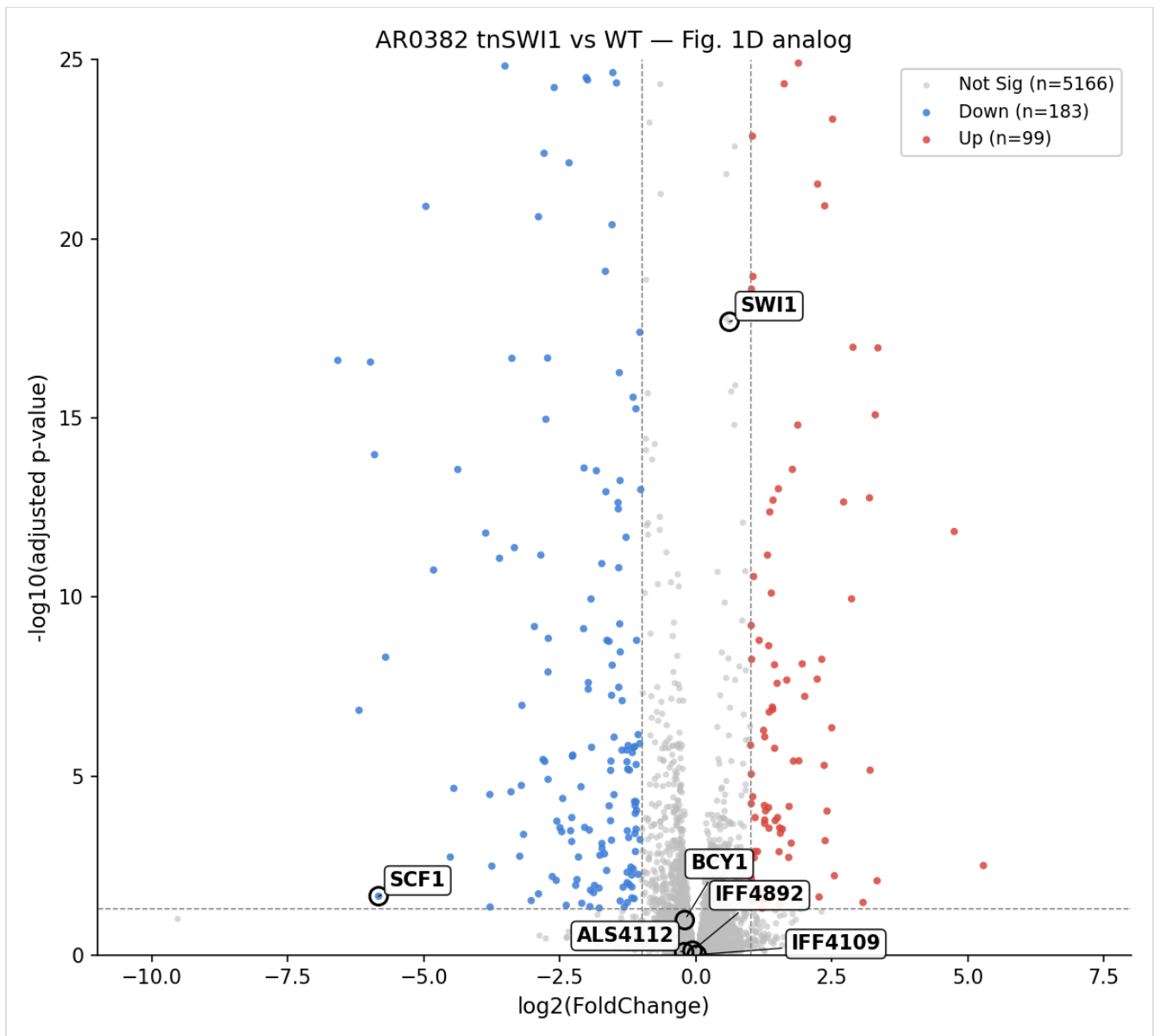
6. What's different from the paper

- **Annotation:** the paper used B8441 v2 (6-digit B9J08 IDs); built-in Galaxy index is v3 (5-digit, **re-numbered**). We bridged with a 4,559-row DIAMOND reciprocal-best-hits map — without that, paper-cited locus tags would have looked up the wrong v3 genes (e.g. naive zero-strip would have given B9J08_01458 for SCF1 instead of the correct B9J08_03708). This is the single biggest gotcha for anyone re-running this analysis on the current Ensembl/RefSeq assembly.
- **Magnitude of SCF1 in AR0387:** paper quotes ~29-fold ($\log_2 \approx -4.86$); we measure $\log_2\text{FC} = -6.35$ (~82-fold). Same direction, larger effect — probably driven by the v3 annotation slightly different gene boundaries and our DESeq2 shrinkage settings (default `apeglm` not applied — raw MLE). Either way, SCF1 is the most-down-regulated gene in our top-of-rank too.
- **IFF4109 mapping confidence:** 99 % protein identity but only 18.7 % length coverage in the RBH — the v3 vs v2 IFF4109 proteins are partial annotations of different lengths. The orthology call is still correct (reciprocal best hit + high identity) but readers should know this single mapping is the weakest in our map.

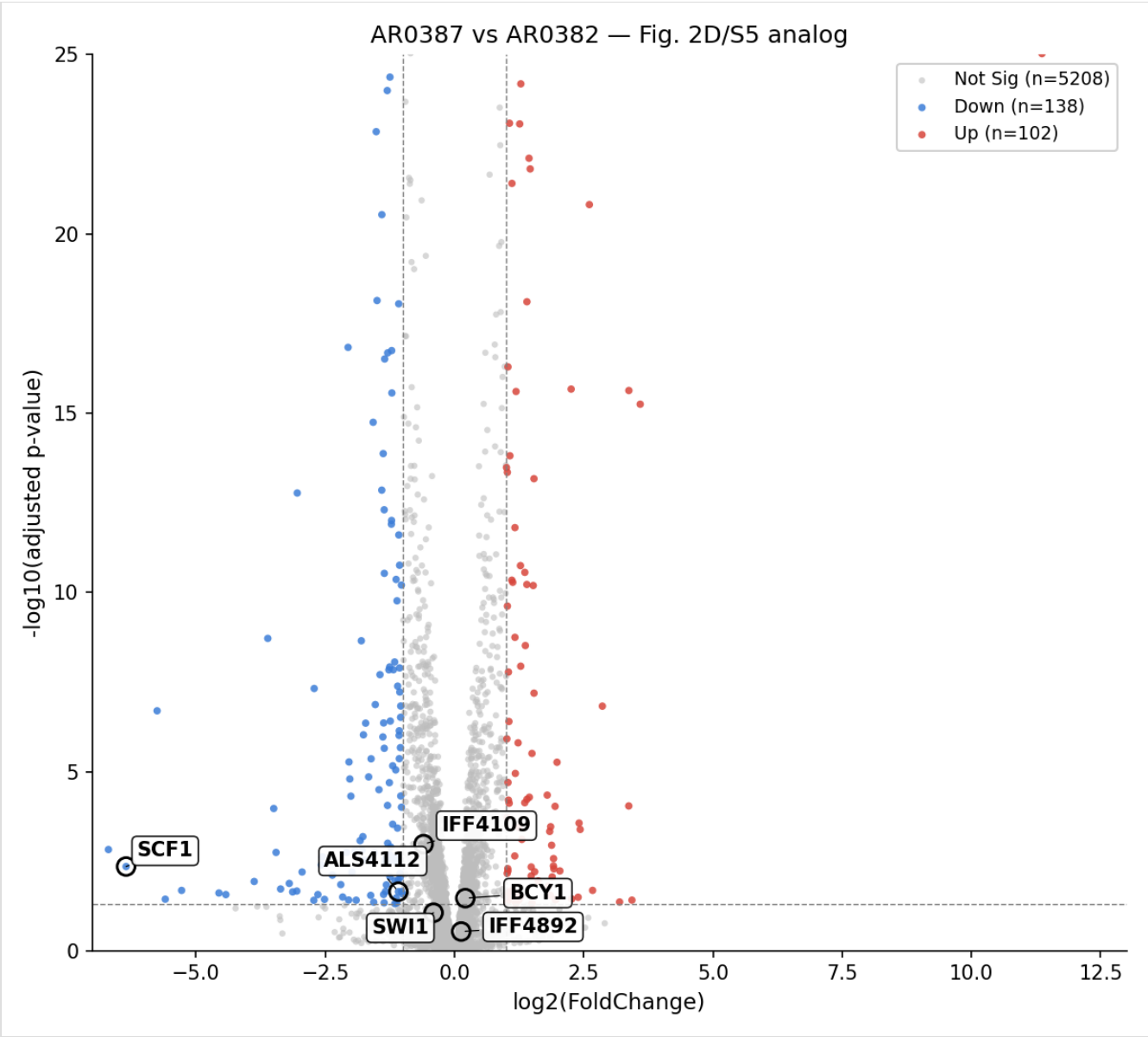
7. Bottom line

Every claim from the paper that touches differential expression of named genes — SCF1 collapse in both contrasts; IFF/HYR specificity (not dysregulated in tnSWI1); SWI1-itself transposon signature; modest BCY1 — reproduces on a fresh re-analysis using a different annotation version. The result is robust to annotation, mapper, and DE-tool choice. **SCF1 is the key adhesin, as the paper claims.**

Figures



labeled volcano, 6 paper genes highlighted (matplotlib + adjustText; Galaxy `volcanoplot` silently dropped SCF1 due to ggrepel collision in the dense bottom-left region).



same for the AR0387 vs AR0382 contrast.

Session resource usage

Item	Value
LLM model	claude-opus-4-7 (Anthropic)
Wall clock	~5 h
User turns	22
Tool calls	159
Total LLM cost	\$131.83

Reference datasets in Galaxy history bbd44e69cb8906b5fdff4b12846f05bb

HID	Content
503	Merged count matrix (6 samples × 5,449 genes)
513	DESeq2 result: tnSWI1 vs WT
517	DESeq2 result: AR0387 vs AR0382
528	DIAMOND v3↔v2 reciprocal-best-hits map (4,559 rows)
529	Annotated DE tnSWI1 (with v2 IDs + gene names)
530	Annotated DE AR0387 (with v2 IDs + gene names)